

Circle Graphs - Examples #1, 2

Tuesday, October 22, 2024 1:50 PM

CIRCLE GRAPHS

A circle graph displays data as sections of a circle. Circle graphs are especially useful to show how a whole amount, such as a budget, is made up of several parts.

In a circle graph, the circle represents the whole amount. Each segment represents a part of the whole. Larger segments represent greater parts of the whole. Data in a circle graph can be displayed as a percent, a fraction, or a decimal. The whole circle can equal 100%, 1, or \$1.00.

IN STATISTICS, this is called a **Pie CHART**.

QUALITATIVE DATA

- categories or attributes

FINDING INFORMATION ON A CIRCLE GRAPH

Example How many students are enrolled as juniors at Grandview Community College?

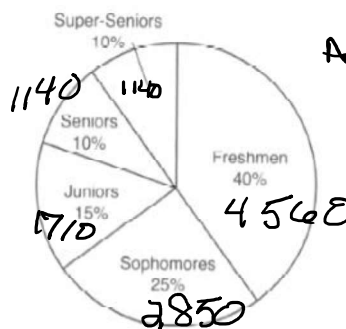
Find the section of the graph labeled "Juniors." It shows that 15% of Grandview's students are enrolled as juniors.

Locate the total enrollment on the graph. Grandview has a total of 11,400 students enrolled during the fall.

15% of 11,400 is 1,710.

1,710 students are enrolled as juniors.

Fall Enrollment at
Grandview Community College
Total Enrollment: 11,400 Students



Adds up to 11400? yes ✓

Use the circle graph above for problems 1 and 2.

1. How many students at Grandview Community College are either seniors or super-seniors?

$$1140 + 1140 = 2280$$

2. Is the combined number of juniors, seniors, and super-seniors greater than or less than the number of freshmen?

$$1710 + 1140 + 1140 = 3990 < 4560$$

less.

Circle Graphs - Examples #3, 4, 5, 6

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Use this circle graph for problems 3-6.

3. Out of each budget dollar, how much did Central High School spend on track?

\$ 0.05 (or, 5¢)

4. Out of a budget of \$20,000, how much money would be spent on softball?

7% of 20,000
 = .07 (20,000)
 = 1400.00
 = \$ 1400

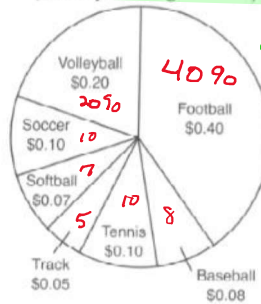
5. The amount spent on football equipment is equal to the amount spent on which three sports?

Football 40% = Volleyball 20%
 Soccer 10%
 Tennis 10%

6. What percent of the equipment budget was spent on baseball?

8%

Approximate Equipment Spending
 Central High School
 (cents per budget dollar)



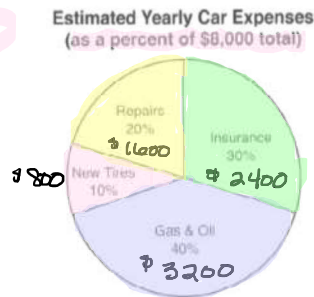
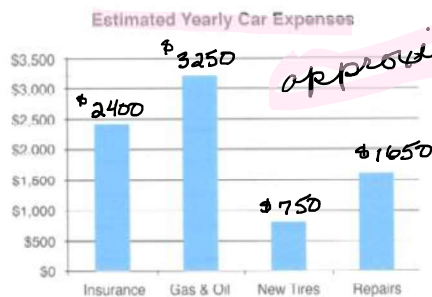
For example,
 40¢ per \$1.00 Budgeted
 \$4 per \$10
 \$40 per \$100
 \$400 per \$1000
 etc

Circle Graphs - Examples #7, 8

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Use the bar graph and circle graph below to answer problems 7 and 8.

The graphs below both show Beverly's yearly car expenses.



7. Name two expenses that combine to equal about one-half of Beverly's yearly budget. Which graph did you use?

total $\approx$ \$8050
Ins + Repairs $\approx$ \$4050

8. **Explain** In which situations is a circle graph most useful? In which situations is a bar graph most useful?

(colored pie charts for Budgets !!)

pie charts : percentages %

bar graphs : descending or increasing order



Circle Graphs - Creating a Graph from Data

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CORE CONNECTIONS: Creating a Circle Graph

In a circle graph, the size of each segment depends on what percent of the total it represents. For example, a segment representing 20% of the whole will be 20% of the entire circle. To create a circle graph, find the number of degrees for each segment.

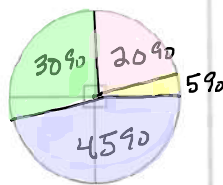
When a circle is divided into fourths, there are four right angles. You know that a right angle measures 90°. Since the total number of degrees in a circle equals the sum of all four angles, there are 360° in a circle: $4(90^\circ) = 360^\circ$

You can find the size of a segment in a circle graph by taking a percent of 360°. Look at the graph of Estimated Yearly Car Expenses. The segment for car repairs is 20%. Since 20% of 360° is 72°, the size of the segment labeled "Repairs" measures 72°.

Complete the table to the right.

A total of 11,000 votes were cast in a recent city election. Use the total to find the percentage of votes for each candidate. Then use the percent to find the number of degrees it would take for each candidate's segment in a circle graph.

	Votes	Percent	Degrees
Kohli	3,300	30%	108°
Sanderson	550	5%	18°
Morales	4,950	45%	162°
Souba	2,200	20%	72°



$$\text{Kohli } 3300/11000 = 33/110 = 30\%$$

$$\text{Sanderson } 550/11000 = 55/1100 = 5\%$$

$$\text{Morales } 4950/11000 = 45\%$$

$$\text{Souba } 2200/11000 = 22/110 = 2/10 = 1/5 = 20\%$$

out of 360°

$$108^\circ + 18^\circ + 162^\circ + 72^\circ = 360^\circ$$

ALWAYS
ADD &
CHECK

QUANTITATIVE (NUMERIC)
DATA

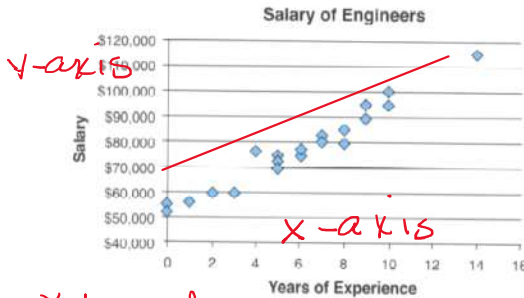
SCATTER PLOTS

A scatter plot is a graph that shows the possible relationship between two different sets of data. Scatter plots are useful in graphing the data from a survey or poll.

The table shows the partial results of a survey. A total of twenty engineers were asked their years of experience in their field and their salary.

The results of the survey are represented on the scatter plot below. In this graph, the horizontal axis represents years of experience. The vertical axis represents salary.

Years of Experience	Salary
0	\$55,000
2	\$60,000
5	\$70,000
10	\$100,000
3	\$60,000
8	\$80,000
6	\$75,000
5	\$73,000



You can find individual points from the survey on the graph, but a scatter plot is meant to show the relationship between the data sets.

Is there a relationship between an engineer's experience and salary? The data on the graph show that the more experience an engineer has, the higher the salary.

∴ as experience increases, salary increases.

Y depends on X !!

You can say there is a correlation between years of experience and salary. You could even identify the type of correlation.

CORRELATION ≠ CAUSATION

lots of experience does NOT guarantee a large salary;
∴ correlation NOT causation.

Correlation or Causation?

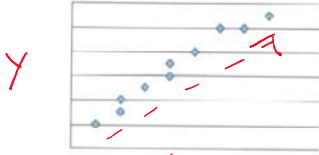
1. Guns in the home & accidental shooting deaths in the home - **Correlation** because a gun in the home does not guarantee a shooting.
2. Exercise & good blood pressure - **Correlation** because even though exercise generally helps lower blood pressure, there could be other confounding variables that prevent the lowering of the blood pressure.
3. Running a red light & getting caught by a red-light camera - **Causation** because the camera catches every car going through an intersection.

Correlation - Examples #5, 6, 7, 8

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A **positive correlation** means that when values in one set of data increase, the values in the other set also increase.

LINEAR



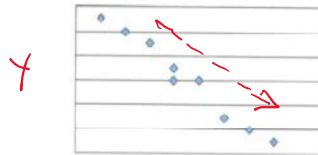
Years of experience and an engineer's salary have a positive correlation.

Sometimes sets of data have no relationship, or **no correlation**.

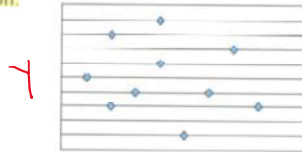
Hat size and the number of pets owned have no correlation.

A **negative correlation** means that when values in one set of data increase, the values in the other set of data decrease.

LINEAR



The number of days absent and a student's grades might have a negative correlation.



other types of correlations:

exponential

logarithmic

cubic

quadratic

A classic example of a logarithmic correlation is the relationship between the **Richter scale and earthquake intensity**, where each whole number increase on the scale represents a tenfold increase in the earthquake's strength;

An example of an exponential correlation would be the **growth of a bacterial population over time**, where the number of bacteria increases rapidly at a

Decide whether the following groups of data would likely have a positive correlation, a negative correlation, or no correlation.

5. Math test score and mile run time

no correlation

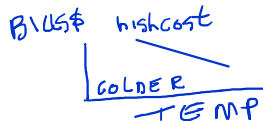
6. Temperature and ice cream sales

positive



7. Temperature and heating bills

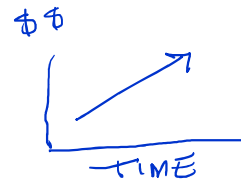
negative



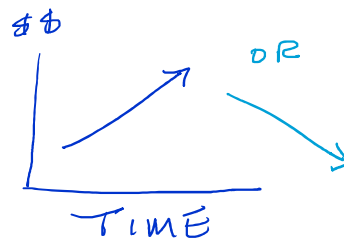
8. Length of investment and interest earned

depends on investment.

1) guaranteed returns as in a bank, positive correlation



2) stock market



Scatter Plots - Examples #1, 2

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Use the scatter plot for problems 1 and 2.

The scatter plot shows the results of a study comparing the maximum distance from which a driver can read a street sign and the age of the driver.

1. Is there a correlation between the two groups of data? If so, is it positive or negative?

NEGATIVE

2. Theo is 35 years old. Based on the results of the study, predict the maximum distance from which he can read a street sign.

~450 ft.



Question: does being older guarantee one's ability to read a street sign decreases?

No !!

if it were, then this would be a CAUSATION Relationship.

Scatter Plots - Examples #3, 4

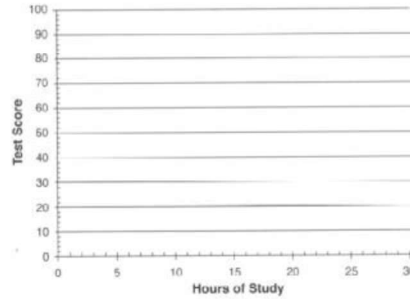
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Use the table and scatter plot below for problems 3 and 4.

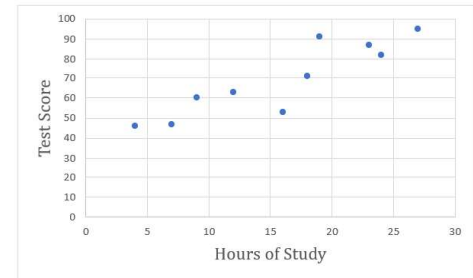
A company requires its employees to take a policy test every two years. Employees can study an online handbook that tracks hours. The results are listed in the table below.

3. Complete the scatter plot using the data in the table.

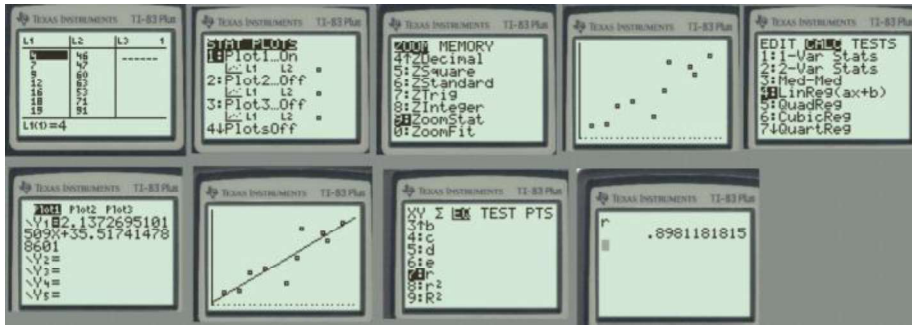
Hours of Study	Test Score
24	82
7	47
16	53
27	95
19	91
9	60
23	87
4	46
12	63
18	71



Hrs	Score
4	46
7	47
9	60
12	63
16	53
18	71
19	91
23	87
24	82
27	95



4. Does the pattern on the scatter plot show a correlation between employees' test scores and hours of study? If so, is the correlation positive or negative? *yes, positive correlation*



Line Graphs - Examples #1, 2, 3, 4

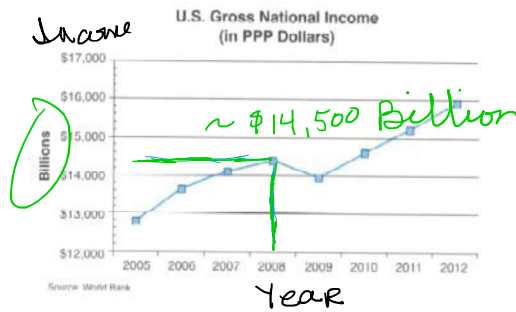
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Quantitative data
Scatterplots & line graphs are usually seen together

LINE GRAPHS

A line graph is a graph that connects the plotted data points with a line. Line graphs are especially useful when showing how a quantity changes over time.

The line graph at the right shows the gross national income of the United States from 2005 to 2012. The gross national income is the total value of goods and services produced by a nation. The dots represent data points. From the graph, you can see that the gross national income has mostly increased over time, with only one period of decrease.



FINDING INFORMATION ON A LINE GRAPH

Example What was the *approximate* gross national income of the United States in 2008?

Step 1

Locate 2008 on the horizontal axis and find the data point directly above it.

Step 2

Read the value on the vertical axis closest to the data point.

The data point is about halfway between 14,000 and 15,000. Note that the values on the vertical axis represent billions of dollars. The gross national income in 2008 was about **\$14,500 billion**.

Use the line graph above for problems 1–4.

1. In which year was the gross national income approximately \$13,600 billion? **2006**

2. By about how much did gross national income increase from 2009 to 2012?

14,000 Billion to 15,800 Billion ≈ \$1,800 Billion

3. In which year did the gross national income increase the most?

Hard to tell, 2011 to 2012 ??

4. In which year did the gross national income decrease?

2008 to 2009

By the way, anyone know what happened in 2008?

